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Chapter 1

Tensor Fundamentals

Chapter References

The references used in this chapter are:

- Simo & Hughes: Computational Inelasticity [?]
- Ted Belyschko: Nonlinear Finite Elements for Continua and Structures [?]

This chapter will cover the tensor essentials to understand tensor algebra applied to solid mechanics.

Example 1.0.1: First test with example

This is an example

Key Concept 1.0.1: Einstein notation

This is a concept

1.1 Index Notation & Einstein Summation

1.2 Stress Tensor Derivation

1.3 Strain Tensor Derivation

1.4 Hand Calculations

Chapter 2

Principal Stressses

Chapter References

The references used in this chapter are:

- Simo & Hughes: Computational Inelasticity [?]
- Ted Belyschko: Nonlinear Finite Elements for Continua and Structures [?]